

Cyber Security Internship Report

Task: Phishing Email Analysis

Objective

The objective of this task is to analyze a phishing email, identify common phishing indicators, and understand the techniques used by attackers to deceive users into revealing sensitive information.

1. Sample Phishing Email

Subject: Urgent: Your Netflix Subscription Has Been Suspended

From: billing@netflix-support-secure.com

To: user@example.com

Email Body:

Dear Customer,

We were unable to process your latest payment for your Netflix subscription. As a result, your account has been temporarily suspended.

To avoid permanent cancellation, please verify your billing information within the next 24 hours by clicking the link below:

Verify Your Account:

<http://netflix-secure-billing-update.com>

If you do not update your payment information within 24 hours, your account will be permanently deactivated.

Thank you for choosing Netflix.

Sincerely,
Netflix Billing Team

2. Examination of Sender's Email Address

Sender Email: billing@netflix-support-secure.com

Findings

- The sender claims to represent Netflix.
- The domain is **netflix-support-secure.com** instead of the legitimate **netflix.com**.
- Attackers often register lookalike domains to impersonate trusted organizations.

Conclusion:

The sender's email address is suspicious and indicates a spoofing attempt.

3. Email Header Analysis

Tool Used: Online Email Header Analyzer (e.g., Google Admin Toolbox or MXToolbox)

Observations

Header Field	Observation
From	billing@netflix-support-secure.com
Return-Path	Different from sender address
Reply-To	Different domain
SPF	Failed
DKIM	Failed / Not Found
DMARC	Failed
Mail Server	Originated from an unknown mail server

Findings

- Authentication checks fail.
- Sender information is inconsistent.
- The email does not originate from Netflix's legitimate mail servers.

Conclusion:

The header analysis strongly indicates a phishing email.

4. Suspicious Links and Attachments

Suspicious Link

<http://netflix-secure-billing-update.com>

Findings

- The URL is not the official Netflix website.
- Uses a deceptive lookalike domain.
- Uses HTTP instead of HTTPS.

Attachments

No attachments were included in this sample.

Conclusion:

The malicious link is intended to steal user credentials.

5. Urgent or Threatening Language

The email uses several pressure tactics, including:

- "Your account has been temporarily suspended."
- "Verify your billing information within the next 24 hours."
- "Your account will be permanently deactivated."

Findings

These statements create fear and urgency, encouraging victims to act without verifying the email.

Conclusion:

Urgency and threats are common social engineering techniques used in phishing attacks.

6. Mismatched URLs

Claimed Service

Netflix Billing Verification

Actual URL

<http://netflix-secure-billing-update.com>

Findings

- The URL does not belong to Netflix.
- The domain attempts to imitate the official website.
- Users may mistake it for a legitimate website.

Conclusion:

The URL is deceptive and intended to redirect victims to a fake login page.

7. Spelling and Grammar Verification

Observations

- No major spelling mistakes.
- Uses a generic greeting ("Dear Customer").
- Contains generic wording and an overly urgent tone.

Findings

Modern phishing emails often have correct grammar to appear authentic.

Conclusion:

The absence of spelling mistakes does not guarantee that an email is legitimate.

8. Summary of Phishing Traits

Phishing Indicator	Status
Fake sender address	✓ Detected
Lookalike domain	✓ Detected
Suspicious URL	✓ Detected
Urgent language	✓ Detected
Threatening message	✓ Detected
Generic greeting	✓ Detected
Email authentication failure (SPF/DKIM/DMARC)	✓ Detected

Phishing Indicator	Status
Malicious attachment	✗ Not Present

Final Conclusion

The analyzed email exhibits multiple characteristics commonly associated with phishing attacks. These include a spoofed sender address, a deceptive domain name, a suspicious verification link, urgent and threatening language, generic greetings, and failed email authentication checks. Although the email contains no malicious attachment and has minimal grammatical errors, the combination of these indicators strongly suggests that it is a phishing attempt designed to steal sensitive information such as login credentials or payment details.

Users should always verify the sender's identity, inspect URLs carefully before clicking, avoid responding to unsolicited requests for personal information, and report suspicious emails to their organization's security team.